

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY****Seventh Semester of B. Tech. (EC) Examination****March-April 2018****EC-405.01/405 Digital System Design****Date: 06.04.2018, Friday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

**SECTION-I****Q-1 Attempt the following. [07]**

- a A wrong output signal produced by a defective system is called \_\_\_\_\_. 01
- b \_\_\_\_\_ are high density device (PAL,PLA,ASIC) 01
- c For NAND gate all the input s-a-0 faults and the output \_\_\_\_\_ faults are equivalent. 01
- d The Boolean expression  $(a'+b')(b+c)$  contains static 0 or static 1 hazard? Justify your answer 02
- e Differentiate Between Moore and Mealy FSM. 02

**Q-2(a) Draw Mealy and Moore state graph for sequence 1011. [04]****Q-2(b) Answer any two. [10]**

- (i) Draw a block diagram of a 4-bit serial adder with Accumulator. Explain operation of it and also describe control state graph.
- (ii) Explain Antifuse and SRAM programmable switching technology in detail.
- (iii) Draw and explain the complete design flow for ASIC. Highlight the stages for which it will be different from programmable logic flow in flow chart. Explain the reason of this difference in detail.

**Q-3(a) Using the method of map entered variables, use 4 variable maps to find a minimum sum of product expression. (No need to verify with 5 variable kmap) [04]**  
 $F(A,B,C,D,E) = \sum m(0,4,6,13,14) + \sum d(2,9) + E(m1,m12)$

**Q-3(b) Answer any two [10]**

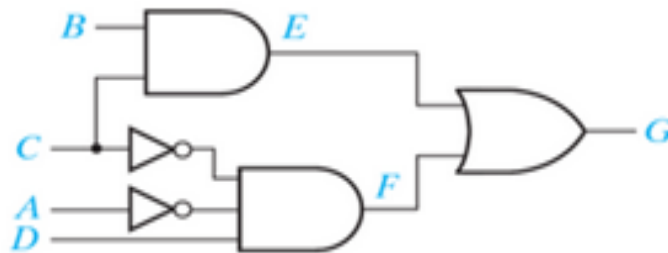
- (i) Which are the main modules of keypad scanner? Explain the working of key scan using state graph.
- (ii) What is programmable logic? What kinds of programmable logic devices are available today? How are they different from one another?
- (iii) Explain the block diagram, flow chart and SM chart for Dice game.

**SECTION II****Q-4 Do as directed. [07]**

- a A static-0 hazard occurs in \_\_\_\_\_circuits and static 1 hazard occurs in \_\_\_\_\_ circuits (OR-AND, AND-OR,AND-AND,OR-OR) 02
- b Covert state graph of 1010(Melay/overlapping) sequence detector to ASM Chart. 03
- c Hardware imperfection is called \_\_\_\_\_.(Error, Defect, Fault) 01
- d EPROM is \_\_\_\_\_programming technique(Non volatile, Volatile) 01

**Q-5(a) What is BILBO? Describe BILBO architecture and its operation? [05]****OR**

- (a) Design a n/w which divide 6 bit divided by 3-bit divisor with suitable example and state graph [05]
- (b) Modify the Figure 1 to eliminate hazards. [04]

**Figure 1(Q-5(a))**

- (c) Design sequential traffic light controller for the intersection of a street “A” and “B” street with state graph. [05]

**OR**

- (c) A sequential network has one input (x) and one output (z). The output  $z=1$  if the total number of 1's received is divisible by 3(Note: 0,3,6, 9,... are divisible by 3).. [05]

**Q-6(a) Compare the volatile and nonvolatile programmable switching technology in detail. [06]**

- (b) What is Fault Dominance? Find out fault dominance class for below mentioned gates. [03]

AND,OR,NAND,NOR

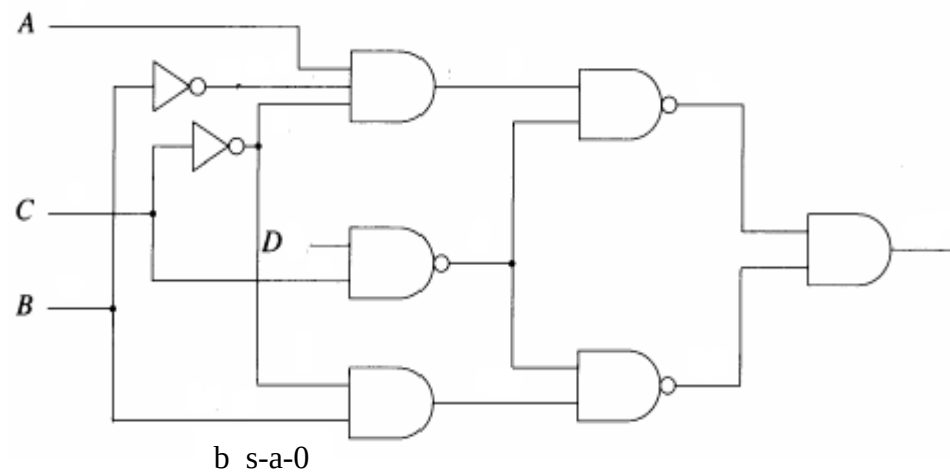
- (c) Draw and explain the SM chart for UART transmitter. [05]

**OR****Q-6(a) Given the polynomial  $P(x) = x^4+x+1$ . [04]**

- (i)Design a LSFR with characteristic polynomial  $P(x)$ .
- (ii)Initial state of the LFSR is “1000” then determines the complete sequence.

(b) For the Figure 2 given below,

[06]



**Figure 2(Q-6(b))**

Find the set of all tests that detect the fault b s-a-0

(c) Explain Non volatile memory. Which programming technology comes under this group? [04]